

Diagnostic reasoning

C/O “chest pain starting 3 days ago”

HPI:

B. H is 31 yrs old female with a PMH of asthma and anxiety who presents to ED complaining of chest pain. She is not known at any cardiology clinic. Pertinent review of history shows that patient was last seen in the ED in October 2024 at Temple for jaw pain after blunt trauma.

Patient C/O chest pain starting 3 days ago and SOB. She says that her chest pain started during episodes of intermittent nausea and vomiting. Her pain is characterized as tightness and aching and is located at her throat, radiating to substernal. She said there is no timing, but each episode of pain lasts about 10-20 minutes. She denies any aggravating factors. She tried taking Tums, rolaids, and pepto bismol, but none of these helped. Associated symptoms are heart palpitations with dizziness.

In the ED, Labs are notable for Na 129, K 2.8, Cl 80, Bicarb 36, BUN 16, Scr 0.57, Ca 9.9, ALP 101, AST 83, ALT 92, WBC 5.8, Hgb 15.9, MCV 88, PLT 248, HS troponin negative. EKG showed sinus tachycardia with rate of 108 with ST depression in II, III, AVF, and V3 - V6. She was taken to the cath lab and to be admitted to the Cardiac PCU for further chest pain workup.

ROS:

- **General:** Reports fatigue and loss of appetite
- **HEENT:** Reports sore throat. Denies Blurred vision, congestion, ear discharge, ear pain, hearing loss, nosebleeds, sinus pain and tinnitus.
- **Respiratory:** Reports shortness of breath. No cough, hemoptysis, sputum production, wheezing, and stridor.
- **Cardiovascular:** Reports chest pain and palpitations. No orthopnea, claudication, leg swelling and PND.
- **Gastrointestinal:** Reports abdominal pain, heartburn, nausea and vomiting. No blood in stool, constipation, diarrhea and melena.
- **Genitourinary:** No dysuria or hematuria.
- **Musculoskeletal:** Denies weakness, joint pain, swelling, stiffness, or back pain.
- **Skin:** Denies rash, lesions, pruritus or significant skin changes.
- **Neurological:** Reports dizziness. Denies tingling, tremors, sensory change, speech change, focal weakness, seizures, loss of consciousness, weakness, and headaches.
- **Psychiatric/Behavioral:** The patient is nervous/anxious.
- **Hematologic/ Lymphatic:** Denies easy bruising, bleeding, or lymphadenopathy

Allergies:

- Penicillin - Hives

PMH: Asthma, Anxiety

Surgical history: NA

Social History: She uses occasional recreational marijuana, social drinking, and smoking 1 cigarette per day.

Family history:

- Maternal Grandmother: Diabetes
- Maternal Aunt: Heart
- Parents: Healthy and alive.

Home Medication:

- Ondansetron (Zofran) 4 mg , 3 times daily PO as needed
- Ibuprofen 600 mg PO every 6 hours as needed for mild pain
- Metoclopramide 10 mg PO every 8 hours as needed for nausea/vomiting
- Omeprazole 20 mg DR PO daily
- Oseltamivir 75 mg PO every 12 hours

Inpatient Meds:

- aspirin chewable tablet 324 mg (324 mg oral Given 1/27/26 1408)

Laboratory Data:

Lab	Units	01/27/26
		1356
SODIUM	mmol/L	129*
POTASSIUM	mmol/L	2.8*
CARBON DIOXIDE	mmol/L	36*
UREA NITROGEN	mg/dL	16
CREATININE	mg/dL	0.57
CALCIUM	mg/dL	9.9
MAGNESIUM	mg/dL	2.1
BILIRUBIN TOTAL	mg/dL	1.1
ALT	U/L	92*

AST	U/L	83*
GLUCOSE	mg/dL	95

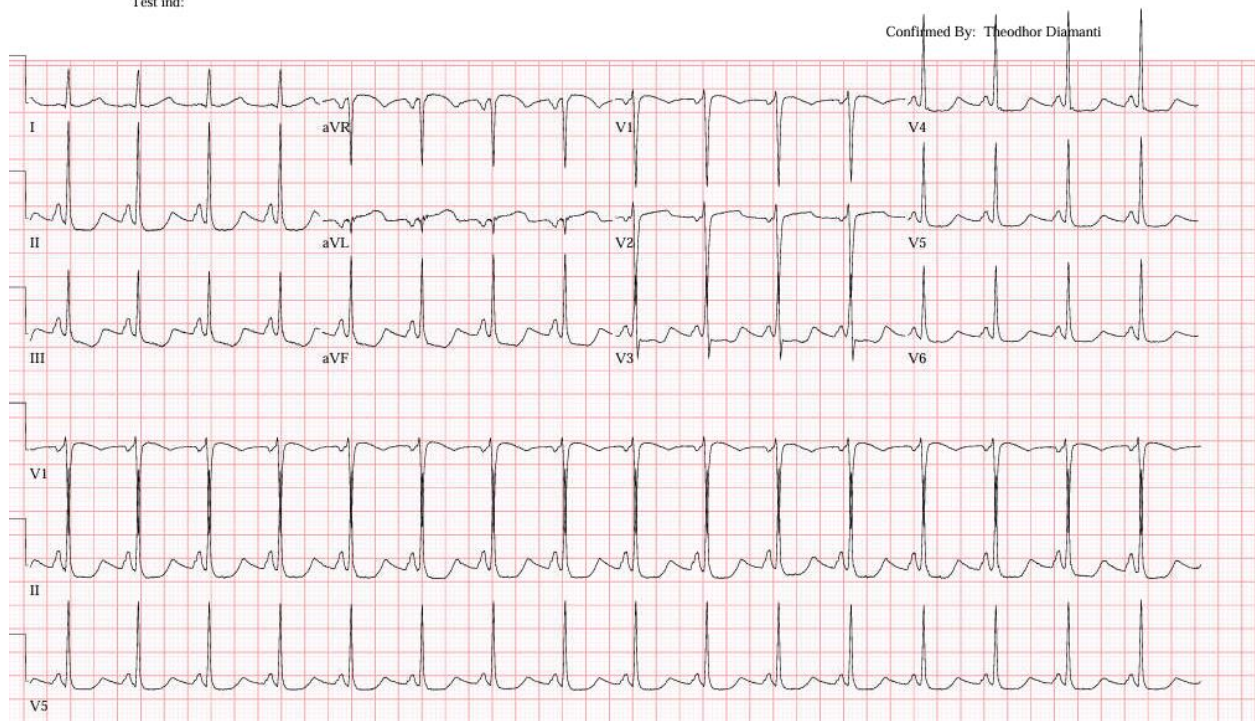
WBC	10³/uL	5.8
HEMOGLOBIN	g/dL	15.9
HEMATOCRIT	%	45
PLATELETS	10³/uL	248

HS Troponin	1/27/2026: Troponin T, High Sensitivity (HS TNT) <6
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Related Images:

- **Electrocardiogram** (1/27/26): Sinus tachycardia , Normal Axis, QRS narrow . There are ST depressions in inferior leads and V3 - V6

Technician:
Test ind:



➤ **CARDIAC CATHETERIZATION (01/27/26):**

Normal coronary arteries in a right dominant system.

HEMODYNAMICS:

LVEDP = 0 mm Hg

No significant peak to peak gradient on pullback across aortic valve

• **CORONARY ANGIOGRAPHY:**

1. Left Main: Large caliber vessel which gives rise to the LAD, ramus and LCx. There is no disease.
2. Left Anterior Descending: Large caliber vessel which gives rise to two diagonal branches and multiple septal perforators. There is no stenosis.
3. Ramus: Large caliber vessel supplies anterolateral wall. There is no stenosis.
3. Left Circumflex: Large caliber vessel which gives rise to three obtuse marginal branches. There is no stenosis.
4. Right Coronary: Dominant, large caliber vessel which gives rise to the posterior descending artery and a small posterolateral branch. There is no stenosis.

- **CT-PE (1/27/26):** Bronchitis. No evidence of pneumonia. No evidence of pulmonary embolus.
- **TTE (1/28/26):** Normal biventricular size and function (LVEF 62%). No significant valvular dysfunction is seen.
- **CXR 01/27/26:** No active disease is seen in the chest No pneumothorax, is seen. Normal cardiomediastinal silhouette.



Objective data:

Vital Signs:

BP: 115/90, **Pulse:** 85 . **Resp:** 14, **Temp:** 97.3 °F (36.3 °C) and **Spo2:** 99% on RA

Wt: 108 lbs

Physical Exam

- **General:** Ill-appearing female, awake and cooperative, in no acute distress.
- **HEENT:** Normocephalic, atraumatic. Scalp without lesions or tenderness. Eyes: PERRLA.EOMI. Conjunctiva pink, sclera anicteric. No discharge. Visual fields grossly intact.Mouth/Throat: Lips moist. Oral mucosa moist without lesions. Dentition in good repair.Oropharynx clear without erythema or exudate. Uvula midline. Tonsils without hypertrophy.

- **Cardiovascular:** Regular rate and rhythm. Normal S1/S2. No murmurs, rubs, or gallops. Peripheral pulses 2+ and symmetric.
- **Lungs:** Clear to auscultation bilaterally. No wheezes, rales, or rhonchi. Respirations unlabored.
- **Abdominal:** Abdomen soft, non-tender, with normal bowel sounds.
- **Musculoskeletal:** : No deformities or swelling. Full ROM in all extremities. Strength 5/5 bilaterally. No joint tenderness.
- **Skin:** Warm and dry.
- **Neurological:** No focal deficit present. She is alert and oriented to person, place, and time.
- **Psychiatric:** Mood and affect appropriate

Summary:

31yrs old female with history of asthma and anxiety presenting with 3 days of episodic chest pain associated with shortness of breath, nausea/vomiting, and palpitations. Initial evaluation showed sinus tachycardia with inferolateral ST segment depressions and negative high sensitivity troponins. Subsequent left heart catheterization demonstrated normal coronary arteries, lowering concern for obstructive ACS.

Differential diagnosis:

Ddx: Non-obstructive myocardial ischemia > (esophagitis/GERD) > Pulmonary embolism

Diagnostic Reasoning:

1- **Non-obstructive myocardial ischemia- Most likely**

Given the patient's ischemic-appearing EKG changes with persistent chest pain despite negative high sensitivity troponins. Importantly, LHC demonstrated normal coronary arteries, making obstructive ACS unlikely. Normal TTE with preserved LVEF (62%) further lowers concern for structural cardiac disease. Additionally, the patient demonstrates hypokalemic, hypochloremic (Na 129, K 2.8, Cl 80), supporting significant gastrointestinal losses from recurrent vomiting, which may contribute to demand ischemia or microvascular dysfunction.

Plan:

- Continue telemetry monitoring.
- BMP daily.
- Maintain K⁺ > 4, Mg⁺⁺ > 2.
- Consider outpatient evaluation for microvascular angina if symptoms persist.
- Risk factor modification counseling (smoking avoidance, lifestyle counseling).
- Cardiology follows up after discharge.

- 1- **Esophagitis/GERD**- is also **likely contributing** given throat to substernal pain radiation, associated nausea and vomiting, and lack of obstructive coronary disease. Partial symptom context and response to acid suppressive therapy support this possibility.

Plan:

- Continue PPI therapy .
- Chloraseptic spray PRN throat discomfort
- Hold home ibuprofen to prevent symptoms worsening.
- Diet and reflux precautions
- Consider GI follow up if persistent symptoms

- 2- **Pulmonary embolism**- **Rule out:**

Was initially considered given chest pain, dyspnea, and tachycardia; however, CT pulmonary angiography demonstrated no pulmonary embolus and oxygen saturation remained normal at 99% on room air, effectively excluding this diagnosis.

- 3- **Electrolyte abnormalities & dehydration risk**- **ongoing**

Electrolyte abnormalities notable for hypokalemia (K 2.8) and mild hyponatremia (Na 129), likely secondary to intermittent Nausea and vomiting and poor oral intake.

Plan:

- KCl 40 mEq IV over 4 hours
- 0.9% Normal Saline IV at 75–100 mL/hr
- Recheck BMP in the morning
- Maintain Mg > 2.0

Global :

- **Code status:** Full Code
- **Diet:** Regular
- **Discharge:** pending until electrolyte correction is complete and symptoms improve .